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Key Points:

- Robust precipitation decreases during summer in response to CO₂ direct radiative effect are identified in central North America
- Climatology and changes of hydrological environment may collectively modulate the regional CO₂ radiative effect on summer precipitation
- CO₂-induced circulation changes could alter the moisture supply to reshape the regional hydrological cycle

Supporting Information:

Supporting Information may be found in the online version of this article.

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The Direct Radiative Effect of CO₂ Increase on Summer Precipitation in North America

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Abstract Precipitation changes in full response to CO₂ increase are widely studied but confidence in future projections remains low. Mechanistic understanding of the direct radiative effect of CO₂ on precipitation changes, independent from CO₂-induced SST changes, is therefore necessary. Utilizing global atmospheric models, we identify robust summer precipitation decreases across North America in response to direct CO₂ forcing. We find that spatial distribution of CO₂ forcing at land surface is likely shaped by climatological distribution of water vapor and clouds. This, coupled with local feedback processes, changes in convection, and moisture supply resulting from CO₂-induced circulation changes, could determine North American hydroclimate changes. In central North America, increasing CO₂ may decrease summertime precipitation by warming the surface and inducing dry advection into the region to reduce moisture supply. Meanwhile, for the southwest and the east, CO₂-induced shift of subtropical highs generates wet advection, which might mitigate the drying effect from warming.

Plain Language Summary How precipitation changes in full response to increased CO₂ remains unclear. Mechanistic understanding of how increasing CO₂ alone changes precipitation can help us better predict future precipitation in full response to CO₂ increase. We find that the precipitation responses are much stronger during summer than winter in North America. During summer, precipitation significantly decreases in central North America in response to the direct CO₂ radiative effect, while in the southwest and the east, precipitation changes are small. There might be stronger longwave radiation forcing near the surface induced by CO₂ increase in arid regions due to weaker masking effect of water vapor and clouds. This, along with local feedbacks, significantly warms the central region. Meanwhile, the direct forcing of CO₂ induces northward shifts in subtropical highs, consequently generating anticyclonic anomalies. These wind anomalies result in drier air advected into central North America, leading to moisture divergence and hence reducing moisture supply. The reduced water supply, coupled with the warming, may decrease precipitation in the region. For the southwest and the east, the anticyclonic anomalies cause wetter air advected into the regions and generate moisture convergence, which increases moisture supply and might mitigate the drying effect caused by CO₂-induced warming.

1. Introduction

Increasing concentration of greenhouse gasses and associated warming have significantly affected regional precipitation as well as the global hydrological cycle (IPCC, 2021; Meehl et al., 2005; O’Gorman, 2015). The well-known “wet gets wetter, dry gets drier (WWDD)” theory provides a first order depiction of large-scale hydrological changes (Chou et al., 2013; Held & Soden, 2006; Liu & Allan, 2013). However, the WWDD pattern does not hold in regions with significant changes in spatial distribution of winds, temperature and humidity (Byrne & O’Gorman, 2015; Greve et al., 2014; Liang & Zhang, 2022b), where such changes can depend on biases in models’ mean climatology (He & Soden, 2016; B. Zhang and Soden, 2019).

To better understand the mechanism of anthropogenic-induced hydroclimate changes, previous studies have decomposed the forcings into multiple factors, including the direct effect of CO₂ and/or aerosols, global mean SST warming, SST warming patterns, plant physiology changes, etc (e.g., Chadwick et al., 2019; Dong & Leung, 2021; He & Soden, 2016; Mutton et al., 2022). Hydroclimate changes in response to these factors vary across regions and seasons. The mean SST warming plays important roles in weakening tropical circulation and shifting the subtropical jet (Chadwick et al., 2017; He & Soden, 2016; Staten et al., 2012). The mean-warming-

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induced jet changes significantly increase winter precipitation in California (Dong & Leung, 2021). SST warming patterns are found essential in determining the spatial patterns of precipitation changes in the tropics (Chadwick et al., 2014, 2017; Zhang, Stier, et al., 2023). However, the bias in SST patterns in coupled model simulations amplifies uncertainties in the corresponding precipitation responses (Chadwick et al., 2017).

Compared to SST warming, the direct effect of CO₂ radiative forcing can be comparable or even dominate under certain circumstances (e.g., Gaetani et al., 2017; Merlis, 2015; Mutton et al., 2022). Therefore, better understanding of how CO₂ forcing directly influences the hydrological cycle can enhance our confidence in regional precipitation projections and help improve understanding of the full CO₂ response. Several studies have explored the mechanisms of CO₂ direct radiative effect. Bony et al. (2013) and Merlis (2015) pointed out that the direct CO₂ effect can weaken the tropical circulations and alter precipitation patterns by reducing net radiative cooling or its spatial gradients. In contrast, Chadwick et al. (2014) demonstrated that the CO₂ direct forcing primarily affects the precipitation amount rather than its regional patterns. Mutton et al. (2022) showed that the direct CO₂ effect can increase regional precipitation by weakening the meridional circulation over North Africa and inducing more moisture convergence in West Africa. Moreover, Richardson et al. (2016) analyzed the quadrupling CO₂ experiment with fixed SST climatology and found that the annual precipitation over most midlatitude land regions decreases due to a warmer and drier troposphere. However, it remains unclear how CO₂ forcing directly influences dynamics and thermodynamics on regional and seasonal scales, as well as how these dynamic and thermodynamic changes, along with their interactions, shape the spatial patterns of seasonal precipitation changes.

Many studies have sought to explore mechanisms underlying future hydroclimate changes at global scale (e.g., Li & O’Gorman, 2020; Pfhal et al., 2017; Zhao, 2022). For regional precipitation changes, North America is one of the extensively studied regions considering the dense population and fragile ecosystems (Dong et al., 2019; Grise, 2022; Liang & Zhang, 2022c; Ting et al., 2021). However, confidence in future precipitation projections remains limited as models are struggling to reproduce historical precipitation trends over North America, especially at the century scale (Knutson & Zeng, 2018; Zhang, Knutson, et al., 2023). Understanding the direct effect of CO₂ on precipitation can help us better understand the full response of increased CO₂ and gain more confidence on future precipitation predictions. Hence in this study, we focus on precipitation response to CO₂ direct effect in North America. We will utilize multiple state-of-the-art global atmospheric models to identify and extract the robust signal of seasonal precipitation changes in North America in response to CO₂ direct effect, based on evaluations of the interannual variability and intermodal spreads. The relative roles of dynamic and thermodynamic changes for the CO₂-induced hydroclimatic responses will be investigated with the aid of moisture budget analysis. The paper is organized as follows. Section 2 describes the models, observational data, and analysis methods. Section 3 describes the analysis results. Section 4 is the summary and discussion.

2. Data and Methods

We utilized two sets of global atmospheric model simulations to analyze hydrological changes over North America. The first set is from the high resolution GFDL atmospheric model version 4.0 (C192AM4) (Zhao et al., 2018a, 2018b, Zhao, 2020). The C192AM4 utilizes a cubed sphere dynamical core at a horizontal resolution of ~50 km (Zhao, 2020, 2022). It is the model used for GFDL’s participation in the CMIP6 HighResMIP Project. AM4 has been widely used in the hierarchy of GFDL models: climate model CM4 (Held et al., 2019), Earth system model ESM4 (Dunne et al., 2020), and seamless system for prediction and Earth system research SPEAR (Delworth et al., 2020). The second set is from the latest Cloud Feedback Model Intercomparison Project (CFMIP; Webb et al., 2017). CFMIP is a subset of the Coupled Model Intercomparison Project (CMIP; Eyring et al., 2016), focusing on cloud feedback, radiative forcing, and climate sensitivity. Here we utilized 13 CFMIP models that provided quadrupling CO₂ (4xCO₂) simulations (Table S1 in Supporting Information S1). The 13 CFMIP models have various horizontal resolutions and are interpolated to 1° × 1° resolution to derive the multi-model mean.

All models include two sets of specified SST simulations: one to study the influence of 4xCO₂ alone (with unchanged SSTs), and the other to study the impact of a +4 K globally uniform SST change alone (P4K, with unchanged CO₂). In each case these simulations are compared to current climate control simulations (CNTL). It should be noted that we only conducted +2 K global SST warming (P2K) simulation using the C192AM4, and hence we simply doubled the responses in the P2K simulation to approximately emulate the effect of the P4K

simulation. The CNTL, 4xCO₂ and P4K simulations were conducted from 1979 to 2014 with prescribed observed monthly SSTs, sea ice concentrations, greenhouse gasses and aerosols, which include interannual variability (Zhao, 2022). The long-term climatology in the CNTL, as well as the perturbed 4xCO₂ and P4K simulations, is obtained by averaging model output from 1979 to 2014.

In addition to using monthly output to derive changes in climatological mean state, we utilized 3-hourly output from C192AM4 and daily output from CFMIP models (Table S1 in Supporting Information S1) to diagnose changes in precipitation frequency and intensity. Precipitation in C192AM4 is regridded to the same resolution of CFMIP models at 1° × 1°, as resolution can affect magnitude of daily precipitation extremes (e.g., C. Chen and Knutson, 2008). To quantify how 4xCO₂ changes the precipitation frequency, we calculated thresholds using the CNTL experiment with different percentiles (99th, 90th, and 75th) and applied them to the 4xCO₂ experiment. Precipitation exceeding the 75th threshold accounts for more than 90% of the total precipitation (Liang & Zhang, 2022b). For the *N*th percentile frequency changes:

$$\delta F_{Nth} = \frac{M_{P_{4CO_2} > P_{CNTL-Nth}}}{M_{total}} - \left(1 - \frac{N}{100}\right) \quad (1)$$

Where δF_{Nth} is change in precipitation frequency at which precipitation events exceed the *N*th threshold, *P* represents precipitation amount. $M_{P_{4CO_2} > P_{CNTL-Nth}}$ is the number of precipitation events exceeding a certain threshold and M_{total} is the total number of events in the 4xCO₂ experiment. $P_{CNTL-Nth}$ is the *N*th percentile threshold calculated based upon the CNTL experiment. We also calculated the changes in precipitation intensities using different percentile thresholds (see Methods in Supporting Information S1).

To evaluate how well the models can simulate the current climate, we used the Global Precipitation Climatology Project Version 2.3 (GPCP) monthly precipitation starting from 1979 with a resolution of 2.5° × 2.5° (Adler et al., 2018) and the ECMWF Reanalysis version 5 (ERA5) at a horizontal resolution of 0.25° × 0.25° starting from 1950 (Hersbach et al., 2020). Additionally, we employed moisture budget analysis (see Methods in Supporting Information S1) to quantify the impact of changes in dynamics and thermodynamics on the hydrological cycle (Liang & Zhang, 2021; Seager & Henderson, 2013; Ting et al., 2018).

3. Results

3.1. Precipitation Direct Response to 4xCO₂ Radiative Forcing

As the starting basis, we find that both C192AM4 and CFMIP models can capture the major patterns of precipitation, dynamic and thermodynamic processes during different seasons in North America compared to GPCP observations and ERA5 reanalysis in the control climate (Figure S1 in Supporting Information S1), despite non-negligible biases in certain regions. This is consistent with previous studies (Liang & Zhang, 2022a; Ting et al., 2021).

We now examine the model simulated precipitation changes in response to 4xCO₂ radiative forcing and P4K warming over North America. Note that adding the responses from the 4xCO₂ and P4K experiments can well reproduce the response in a 4xCO₂ + P4K simulation (Held & Zhao, 2011). Hence, it is reasonable to decompose the forcing factors to isolate and comprehend the underlying mechanisms. In summer (June-July-August, JJA), precipitation is projected to decrease in response to the direct CO₂ forcing over most parts of North America in C192AM4 and CFMIP models (Figure 1a). Meanwhile, the precipitation response in winter (December-January-February, DJF) is less significant (Figure 1b). Precipitation overall increases in both JJA and DJF under P4K due to significant moisture increases associated with strong mean warming (Figure S2 in Supporting Information S1; Dong & Leung, 2021). Hereafter, we will focus on precipitation reductions in the JJA to diagnose the robust physical mechanisms of the direct effect of CO₂ on changes in the future hydroclimate.

We note that the robustness of CO₂-forced precipitation changes is spatially inhomogeneous over North America. The drying in central and northwest North America is statistically robust regarding both the interannual variability and model spreads, while the changes in the southwest and the east are very small and thus not statistically significant. The lack of robustness in precipitation changes in the southwest could be partially attributed to the relatively small changes and large interannual variance upon a very dry hydroclimate (Figure S3a in Supporting Information S1). The comparable magnitudes of intermodel and interannual variances, along with the absence of

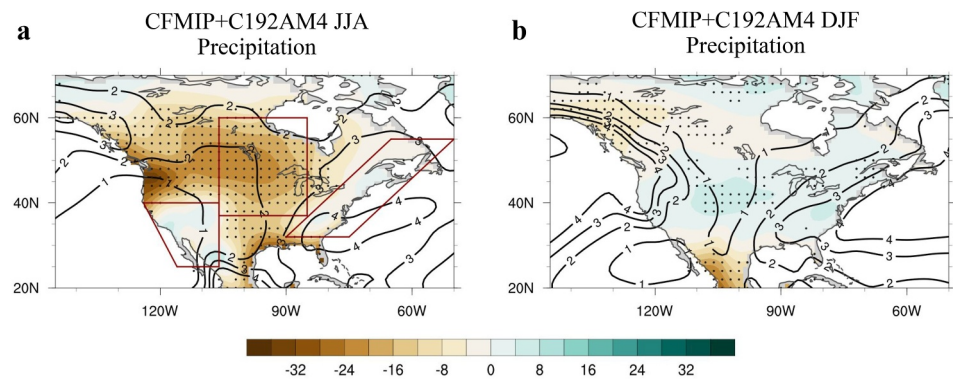


Figure 1. (a) Climatological precipitation (contour, mm/day) in CNTL experiment (1979–2014), and its percentage changes (colors, %) in 4xCO₂ experiment (1979–2014) for C192AM4 and 13 CFMIP models in JJA. (b) Is the same as (a), except for DJF. Stippling indicates that at least 80% of the models agree on the sign of changes. The red parallelograms approximately represent the central, the southwest, and the east North America.

a consistent mechanism (e.g., dynamics changes, thermodynamic changes) to explain the intermodel spreads (not shown here), indicate that the discrepancies among the models in both the southwest and the east are likely attributable to the internal climate variability (Figure S3 in Supporting Information S1; Jain et al., 2023).

Consistent with the decrease in JJA mean precipitation, the JJA precipitation frequency across different percentile thresholds (see Methods) also decreases in central North America (Figure S4 in Supporting Information S1). Changes in precipitation intensity differ among different percentiles if we apply the percentile thresholds from CNTL to the 4xCO₂ experiment (Methods and Figure S5 in Supporting Information S1). Especially, the precipitation intensity of extreme events (above the 99% threshold from CNTL) is generally unchanged (Figures S5a, S5b in Supporting Information S1), even as these events become less frequent in the drier 4xCO₂ climate. In contrast, we observe significant intensity reduction by simply comparing the percentile precipitation between the CNTL and 4xCO₂ experiments in the region (Figure S6 in Supporting Information S1), as the percentile thresholds are generally lower in the drier 4xCO₂ climate.

3.2. Physical Processes for the CO₂-Forced Precipitation Changes

To understand why the CO₂-forced precipitation changes occur in summer, we analyze the two source terms in the moisture budget equation (Methods in Supporting Information S1): surface evaporation and atmospheric moisture convergence. The direct CO₂ effect overall weakens both moisture sources in most parts of North America (Figure 2). Changes in evaporation are well co-located with precipitation changes and exhibit significant decline in both the central and central-southern regions (compare Figures 1a and 1ure 2a). Total moisture convergence has a slightly different pattern, showing decreases across most regions in the north, and some increases in the southwest and the east (Figure 2b). Figure 2c shows that the soil moisture in the top layer of the column decreases in central North America, consistent with reductions in precipitation and evaporation, demonstrating a well-coupled hydrological weakening in the region (Zhou et al., 2023).

3.2.1. Local Radiative Processes

How does the CO₂ direct radiative forcing induce these hydrological changes in JJA? We found that precipitation decreases in central North America are associated with reduction in relative humidity (Figure 3a). The reduced relative humidity near the surface might suppress convection and lead to reductions in both mean precipitation and frequency of extreme precipitation (Williams & O’Gorman, 2022). This reduction in relative humidity could be attributed to warming at the surface, accompanied by the feedback between the atmospheric temperature and humidity through cloud and sensible heat flux, which will be discussed later. Warming increases the atmospheric holding capacity of water vapor and decreases the relative humidity if there is limited water vapor supply. As shown in Figure 3b, the near-surface specific humidity in central North America slightly decreases and thus there is significant reduction in the relative humidity in the region. The strong near surface warming in the region is associated with direct radiative forcing induced by CO₂ increase. Figure 3c shows that there is spatial correlation between changes in surface downwelling longwave radiative flux and climatological relative humidity. The

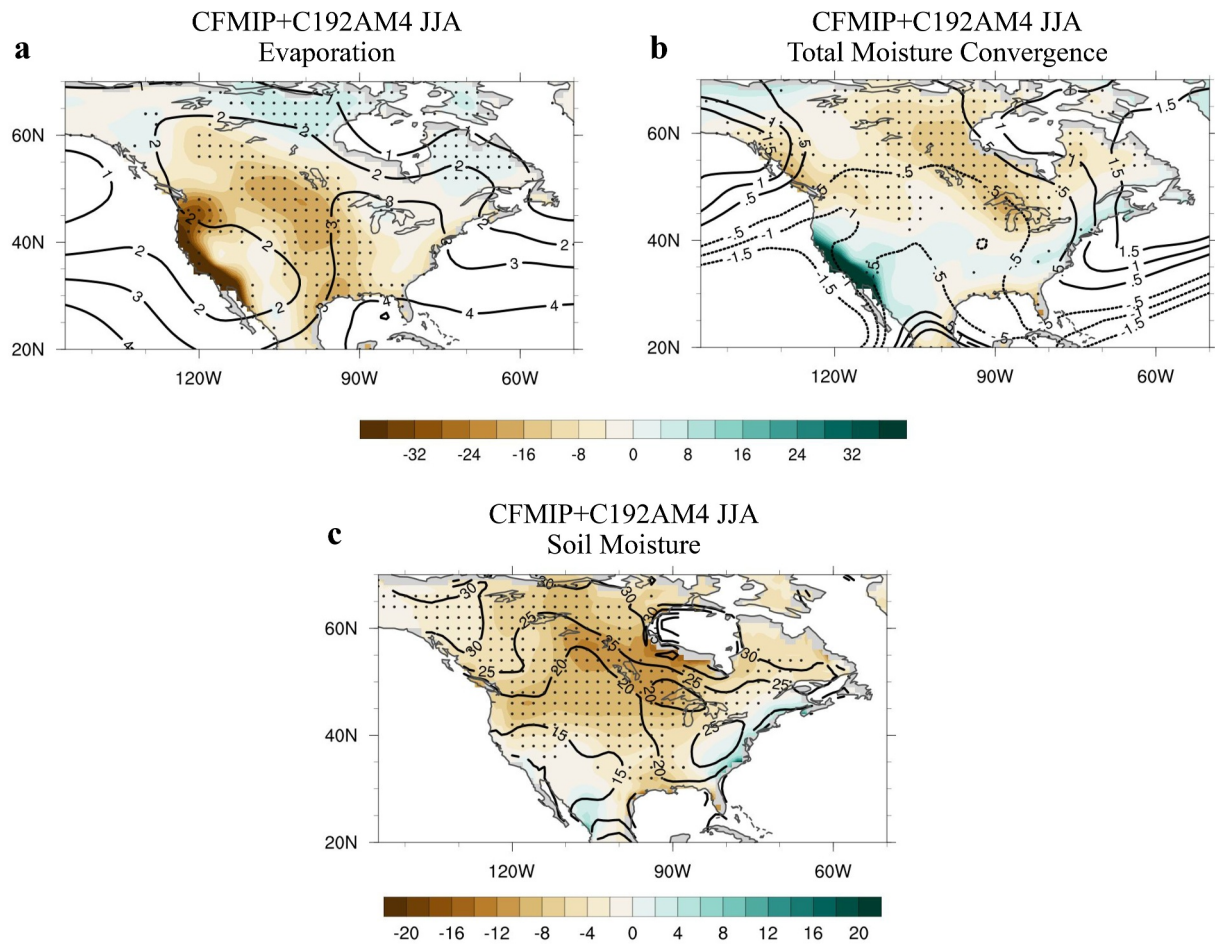


Figure 2. (a) Climatology (contour, mm/day) in CNTL experiment (1979–2014), and percentage changes (colors, %) of evaporation in the 4xCO₂ experiment (1979–2014) in JJA for C192AM4 and 13 CFMIP models. (b) Same as (a), but for total moisture convergence (precipitation minus evaporation). Changes of evaporation in (a) and total moisture convergence in (b) are scaled by climatological JJA precipitation. (c) Climatology (contour, kg/m²) and percentage changes (colors, %) of moisture in the upper portion (10 cm layer) of the soil column. Stippling represents more than 80% of the models agreeing on the change signs.

increase of downwelling longwave radiative flux near the surface is larger in regions with drier climatology (e.g., central and southwestern North America), consistent with the previously identified CO₂ masking effect by clouds and high humidity (Chen et al., 2023; Merlis, 2015). Larger atmospheric water vapor content or a higher total cloud amount in relatively wet regions (e.g., eastern North America) could mask out the CO₂ radiative effect, leading to less radiative forcing reaching the near surface (Figures S7a and S7c in Supporting Information S1). The clear-sky longwave radiative forcing shows similar spatial distributions to those of the all sky conditions (Figure S8a in Supporting Information S1), suggesting that the regions with lower relative humidity generally have less cloud to mask out the CO₂ radiative effect (Figure S7c in Supporting Information S1; Merlis, 2015). The increasing downwelling longwave radiative flux near the surface may induce significant warming across North America (Figure 3a).

In addition to the near surface, the lower troposphere also demonstrates consistent warming responses to the direct CO₂ forcing (Figures S9c and S9d in Supporting Information S1). The 850 hPa warming could be related to the enhanced longwave radiative heating and turbulent heating, which induces stronger downward longwave radiation and thereby amplifying the radiative responses to the direct CO₂ forcing at the surface (Clark & Feldstein, 2020). This warming in conjunction with the slightly decreased 850 hPa specific humidity (Figure S9b in Supporting Information S1), could lead to significantly reduced relative humidity at 850 hPa in central North America (Figure S9a in Supporting Information S1).

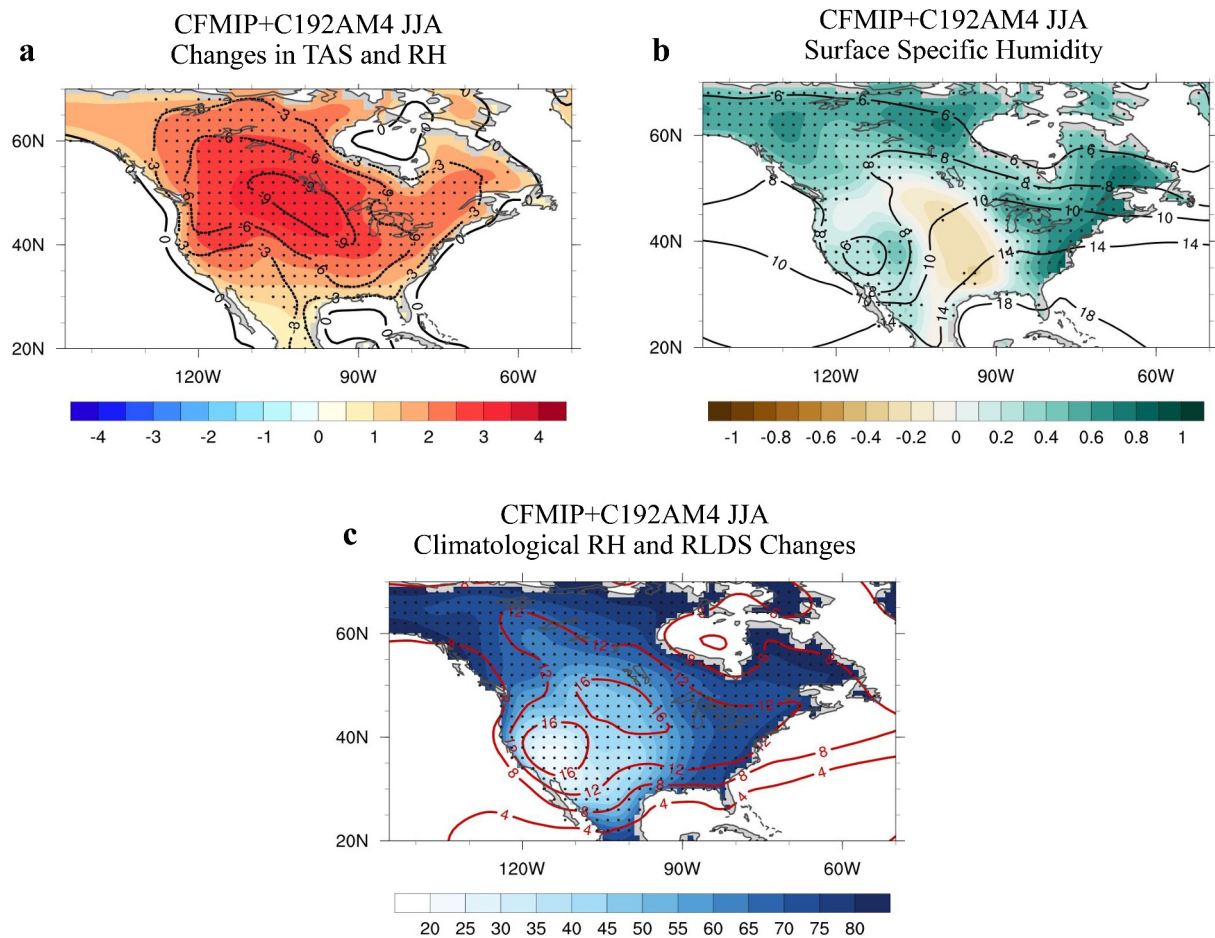


Figure 3. (a) Changes in surface air temperature (TAS) (colors, K) and surface relative humidity (contours, %) in 4xCO₂ experiment (1979–2014) for JJA using C192AM4 and 13 CFMIP models. (b) Climatology (contours, g/kg) in CNTL experiment and changes (colors, g/kg) in 4xCO₂ experiment of surface specific humidity for the C192AM4 and CFMIP models. (c) Climatological surface relative humidity (RH) (colors, %) and changes in surface downwelling longwave radiation (RLDS) (contours, W/m²) in 4xCO₂ experiment using the C192AM4 and CFMIP models. Stippling represents at least 80% of the models agreeing on the sign of changes.

The pattern of climatological relative humidity does not match the warming and precipitation change patterns (compare Figures 1a, 3a and 3c), particularly noticeable in the southwest, suggesting that local radiative processes alone are not the sole mechanisms working. This is further suggested by the increased total moisture convergence in the southwest and the east (Figure 2b). Additionally, this decrease in precipitation is accompanied by reductions in total cloud cover, which might subsequently induce more surface warming (Figure S7c in Supporting Information S1). Reduced evaporation associated with precipitation reduction may also increase the surface sensible heat flux and further warm up the lower atmosphere (Figure S9e in Supporting Information S1). The reduced evaporation could also decrease the available moisture for further precipitation. Such positive feedback could make the warming and drying in central North America even stronger than that of the climatologically driest southwest (Figure 3a). The relatively smaller temperature increase in the southwest and the east could also be related to the ocean-land contrast as the SSTs are fixed in the 4xCO₂ experiment. The heat flux exchange between ocean and land might ease the warming along the coastal regions. Further investigation is needed to verify this possible mechanism.

3.2.2. Non-Local Circulation Changes

Besides the local processes (CO₂ radiative forcing and temperature-humidity feedbacks) as described above, the non-local moisture advection also plays an important role in modulating the pattern of moisture changes and thus the regional precipitation changes. In central North America, the decreased moisture content works with the warming to decrease precipitation, while in the southwest and the east, the increased moisture content counteracts

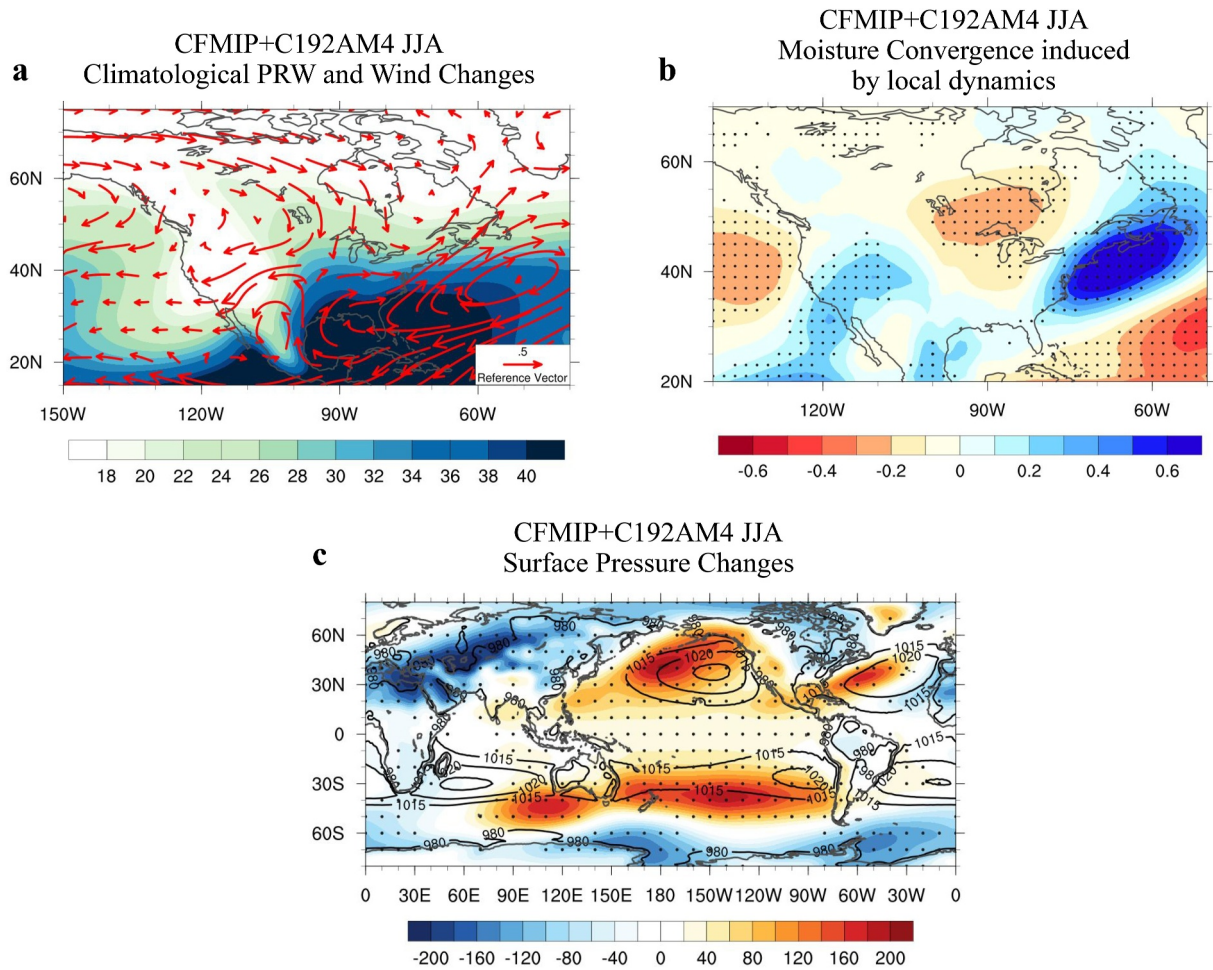


Figure 4. (a) Climatological precipitable water (PRW) (colors, kg/m^2) in CNTL experiment (1979–2014), and changes in vertical-integrated wind weighted by specific humidity (vectors, m/s) in $4\times\text{CO}_2$ experiment (1979–2014) for JJA in C192AM4 and 13 CFMIP models. (b) Changes in moisture convergence (mm/day, blue indicates convergence while red represents divergence) induced by the wind changes in conjunction with the climatological moisture distribution as shown in (a) for C192AM4 and CFMIP models. (b) Is referred to as local dynamics (see Methods in Supporting Information S1). (c) Climatology (contours, hPa) and changes (colors, Pa) of surface pressure (PS) in C192AM4 and CFMIP models. Stippling is where at least 80% of the models agree on the sign of changes.

the drying effect caused by warming, resulting in small precipitation changes thus not statistically significant compared to the climate internal variability and model spreads (Figure 3b). What causes this contrasting response in water vapor supply to the direct CO_2 forcing? Consistent with Ceppi et al. (2018), we find that the direct CO_2 effect leads to a robust northward shift in the subtropical circulations (Figure 4c). The subtropical high over the North Pacific strengthens and shifts poleward, thereby generating anticyclonic circulation anomalies over northwestern North America (Figure 4a). The North Atlantic subtropical high also exhibits a northwestward shift, which induces anticyclonic anomalies near eastern North America. These wind anomalies are up-moisture-gradient and cause dry advection into central North America and generate moisture divergences, hence reducing the water vapor supply (Figures 4a and 4b). In contrast, for the southwest and the east, the down-moisture-gradient wind anomalies induce moist advection accompanied by moisture convergences, providing abundant water vapor to these regions (Figures 4a and 4b). The increasing moisture content might ease the relative humidity reduction caused by CO_2 -induced warming and lead to small precipitation changes in both the southwest and eastern North America (Figures 1 and 3). Especially for the southwest, the CO_2 radiative forcing near the surface is significant due to the dry climatology (Figure 3c, Figure S7 in Supporting Information S1), but the warming and precipitation responses are smaller than those in central North America, which could be due to the increased moisture supply associated with the circulation changes.

4. Summary and Discussions

We have demonstrated the mechanisms underlying precipitation changes in response to the direct CO₂ effect during summer in North America. The lower troposphere warming and relative humidity reduction could be modulated by the combination of climatological hydrological environment (e.g., water vapor and clouds), local feedback processes, and circulation changes. In relatively dry central North America, increasing atmospheric CO₂ concentration could produce significantly warmer land surface due to less water vapor and clouds overlapping with CO₂ direct forcing. Meanwhile the differential warming may alter atmospheric circulation and reduce moisture supply by inducing dry advection into the region. This could result in decreased relative humidity and might suppress the convection, which makes the region “dry gets drier.” Conversely, in the southwest and the east, the drying effect from CO₂-induced warming is likely counteracted by the enhanced moisture supply associated with the northward shift of the subtropical highs. Thus the dry southwest does not get drier (Liang & Zhang, 2022a).

Other than the changes in winds and associated dry (wet) advections, other physical processes also partially contribute to the pattern of moisture changes. Consistent with Ting et al. (2021), weakened transient eddies related to the reduced relative humidity and the southwestward wind anomalies cause moisture divergence in northern North America (Figure S10d in Supporting Information S1). For the southwest, enhanced low-level wind convergence causes moisture convergence, which is likely associated with the maximum jet weakening (Figures S10c and S11 in Supporting Information S1; Liang & Zhang, 2022a). The jet weakening in the relatively south latitudes might be associated with the maximum CO₂-induced warming in central North America (Figure 3a). As for the east, increased low-level wind convergence leads to moisture convergence (Figures S10c and S11 in Supporting Information S1), which might be related to the CO₂-induced poleward shift of storm tracks (Chang, 2013; Grise & Polvani, 2014). The northwest is an exception with relatively wet climatology but still undergoes significant drying, which is likely due to the dominant effect of subsidence anomalies associated with the enhanced North Pacific subtropical high (Figure 4c). The relatively large transient moisture divergence in the southwest and the east is likely related to weakening of the dynamical amplitude of eddies (Figures S10d and S11 in Supporting Information S1; Seager et al., 2014; Liang & Zhang, 2021). Further studies are still needed for better understanding on the mechanism of how CO₂ drives the subtropical circulation changes.

Except for the changes in thermodynamics and dynamics, climatological precipitation might play a role in shaping the precipitation changes. In the relatively dry central region, reduced precipitation could lead to less latent heat flux and more sensible heat flux accompanied by reductions in evaporation, which could subsequently reduce moisture supply for precipitation and make “dry gets drier.” While in relatively wet eastern North America, surface heat fluxes may be less sensitive to reduced precipitation, suggesting that evaporation and sensible heat flux may not be largely influenced, therefore evaporation does not decrease to reduce further precipitation. Further research is needed to verify these possible linkages between precipitation, evaporation and surface heat fluxes.

Changes in hydrological cycle in response to CO₂ increase in DJF are weaker than JJA, mainly manifested in small changes in precipitation and evaporation (Figure S12 in Supporting Information S1). The CO₂-induced longwave radiative forcing is weaker due to the colder climatology in winter than summer, leading to smaller warming responses and relative humidity reduction (Figure S13 in Supporting Information S1). Moreover, mechanisms for precipitation in DJF differ significantly from those in JJA, as they are closely tied to large-scale synoptic scale events, thus the direct CO₂ radiative effect is minor compared to the climatological precipitation mechanism for changes in DJF precipitation (Gao et al., 2014; Miyawaki et al., 2022). Further investigation is needed to elucidate the detailed mechanisms underlying precipitation changes in DJF.

It is noted here that the global atmospheric models demonstrate a systematic dry bias in central North America during summer (Lin et al., 2017), that is likely related to the uncertainties in cumulus parameterizations or irrigations representation in the models (Qian et al., 2020; Sun & Liang, 2023). However, we did not find correlation between the drying trends and the climatological dry bias in the region. This suggests that the changes in summertime hydroclimate in response to the CO₂ direct effect is driven by robust physical mechanisms.

In this study we focused on the CO₂ radiative effects on the hydrological cycle within the lower atmosphere, which contains the majority of atmospheric moisture. As stated in the introduction, CO₂ forcing can influence the

hydroclimate in various ways across different regions and seasons. Hence, a comprehensive quantification of the aforementioned mechanisms is necessary to bolster confidence in future predictions of hydroclimate changes.

Data Availability Statement

- The C192AM4 and CFMIP simulation data can be accessed from <https://esgf-node.llnl.gov/search/cmip6>
- ERA5 data can be downloaded at <https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5>
- GPCP data is available at <https://climatedataguide.ucar.edu/climate-data/gpcp-monthly-global-precipitation-climatology-project>
- The GFDL AM4 model code is available at <http://data1.gfdl.noaa.gov/nomads/forms/am4.0/>.

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